TEXORA · LECTURE SHEET

Process Control in Winding

Winding is the last chance to remove faults before yarn becomes someone else's fabric. Control here means winding efficiency, clearer settings matched to the fabric the yarn is sold for, splice quality, and package quality — each with its routine measurement.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 9 · SPECIALIZATION — YARN MANUFACTURING

1. Winding efficiency

An automatic winder's theoretical output is fixed by winding speed and yarn mass per metre; its real output is cut by clearer cuts, splicing cycles, cop changes and doffing. Efficiency is measured per drum over a shift:

$$\text{Efficiency (\%)} = \text{actual production} / \text{theoretical production} \times 100$$

WORKED EXAMPLE — DRUM EFFICIENCY AND CLEARER ACTIVITY

Yarn Ne 30 (0.01968 g/m); winding speed 1250 m/min; shift 480 min.

Theoretical length = 1250 × 480 = 600,000 m. Theoretical production = 600,000 × 0.01968 = **11.81 kg** per drum per shift.

Actual production weighed = 8.27 kg → efficiency = 8.27/11.81 × 100 = **70.0%**.

Why the drum stood still: clearer cuts. At 90 cuts per 100 km, and with 1 kg of Ne 30 containing 1000/0.01968 = 50.8 km of yarn, cuts per kilogram = 0.9 × 50.8 ≈ **46 cuts/kg** — each cut costs a splice cycle of several seconds plus re-acceleration. A falling efficiency with unchanged settings therefore usually means the yarn arriving has more faults: the winder is reporting on the ring frames.

2. Clearer settings — matched to the end use

Electronic clearers classify faults by length and diameter increase and cut those outside the setting curve. The same yarn needs different settings for different fabrics: a fault invisible in a denim weft is a defect in a fine shirting warp. Settings are therefore part of the product specification, issued per order, not left to habit.

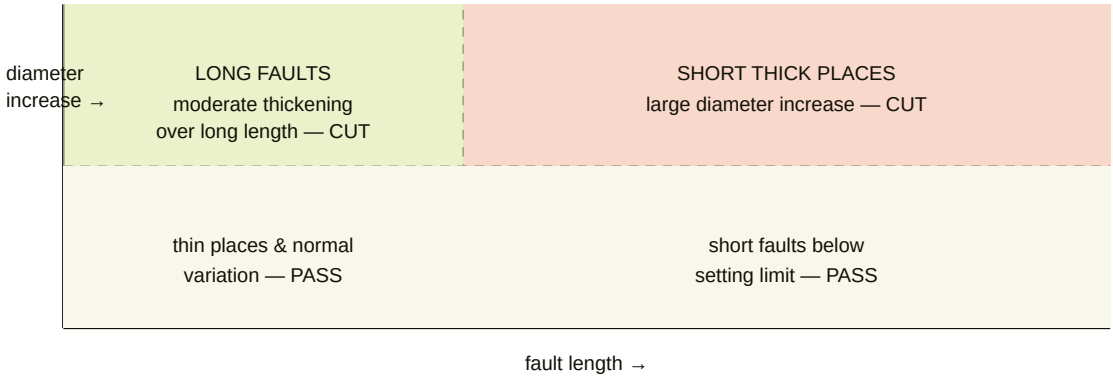


Figure 1 — Clearer logic in schematic: a fault is cut when its combination of length and diameter increase crosses the setting boundary (upper zones). Tightening settings moves the boundary down and left — and raises cuts per kilogram.

FAULT CLASS	TYPICAL SETTING BASIS	EFFECT OF TIGHTENING
Neps / short thick places	Diameter increase % over a few mm–cm	More cuts; cleaner yarn; lower winding efficiency
Long thick places	Moderate increase over 8–40 cm+	Catches slubs and roving faults that survive to fabric
Long thin places	Diameter decrease % over length	Protects warp yarns against weak-place breaks
Count faults (very long)	Small deviation over metres	Removes wrong-count cops mixed into supply

3. Splice quality

Every cut ends in a splice, so clearing creates the defect it must not become. Splice control measures two quantities on laboratory samples: **strength retention** — splice breaking force as a percentage of the parent yarn, targeted at roughly 85% or better — and **appearance/diameter** of the splice, which must pass the clearer itself and stay invisible in fabric. Splicer air pressure, untwisting and overlap settings are verified per drum group; a drum producing weak splices is found by sampling splices drum by drum, not by averaging the department.

4. Package quality

- **Density and build:** cones wound too hard damage yarn and dye unevenly; too soft slough off in transport. Density is checked by weighing cones of known dimensions.
- **Stitches and patterning:** crossed or trapped ends (stitches) on the cone surface cause breaks at warping; pattern zones from the anti-patterning system failing are logged per drum.

- **Length accuracy:** length-controlled doffing keeps cone lengths matched for creeling; checked by the cone-length records against the YM-3207 calculation (a 2.2 kg Ne 30 cone = 111,770 m).
- **Contamination:** coloured fibres, oil stains and mixed lots are final-inspection items — the cone is the product the customer opens first.

Glossary

Clearer — the sensing head that measures yarn diameter continuously and commands a cut when a fault exceeds settings. **Cut** — a clearer-initiated break removing a fault, followed by splicing. **Splice** — the knotless air-joined overlap replacing a knot. **Strength retention** — splice strength as % of parent yarn strength. **Drum** — one winding position of an automatic winder. **Slub** — a long, soft thick place in yarn. **Patterning** — ribbon-like winding coils that form when traverse and rotation synchronise, disturbing unwinding.

Key points

• Efficiency example: theoretical 11.81 kg/drum/shift, actual 8.27 kg → 70.0%; cuts at 90/100 km mean about 46 interruptions per kg of Ne 30.
• Falling winding efficiency is usually a report on ring-frame yarn quality, not a winder fault.
• Clearer settings belong to the product specification; tightening them trades efficiency for cleanliness.
• Splices are controlled on strength retention (≈85%+) and appearance, sampled drum by drum.
• Package density, build and length accuracy are final product controls, checked before despatch.

Practice

1. A winder runs Ne 20 (0.02952 g/m) at 1100 m/min for a 480 min shift and produces 9.6 kg per drum. Calculate theoretical production and efficiency.
2. Clearer cuts rise from 60 to 130 per 100 km after a mixing change, with settings untouched. State the correct interpretation and the two departments you would inform.
3. Why is splice strength sampled per drum rather than as a department average?

Answer key

1. Theoretical length = 1100 × 480 = 528,000 m → production = 528,000 × 0.02952 = **15.59 kg**. Efficiency = 9.6/15.59 × 100 = **61.6%**.
2. The yarn itself has become faultier — the clearer is a measuring instrument and its cut rate is data. Inform ring-frame spinning (fault generation: drafting, roving supply) and mixing/blowroom (nep and trash load), with the fault-class distribution attached so each can see its own signature.

3. Because a splicer fault is usually local — one drum's air supply, nozzle or setting. A department average dilutes one bad drum among many good ones and passes it; per-drum sampling finds and fixes the unit.
